Complex Number 18

1. Use De Moivre's theorem with $n = 2$ to show that $\cos 2\theta = \cos^2\theta - \sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$.Hence show that $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.
2. Express $z = 4\sqrt{2}(1 + i)$ in modulus/argument form. Hence find the two square roots of z and mark their representations on an Argand diagram.

3. a. On an Argand diagram draw a sketch to indicate the locus of z if $\arg(z - 2 + i) = \frac{2\pi}{3}$.

   b. On an Argand diagram graph the region which simultaneously satisfies $1 \le |z - i| \le 2$ and $\mathrm{Im}[z] > 0$.
4. Find the Cartesian equation of the locus of z, given that $z^2 - (\bar{z})^2 = i$.
5. On an Argand diagram P, representing the complex number z, is a point in the first quadrant and $|z| = 2$. If $\mathrm{Arg}\, z = \theta$, find, in terms of θ, expressions for :
   a. $\arg z^2$
   b. $\arg(z - 2)$

6. a. If $|z_1 + z_2| = |z_1 - z_2|$. Find the value of $\arg\left(\frac{z_1}{z_2}\right)$

   b. If $z_1 = 2 + 3i$, find z_2.
7. If ω is a complex cube root of unity, evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$.
8. If z is a complex number such that $z + \frac{1}{z}$ is real, Prove that $Im(\bar{z}) = 0$ or $|z| = 1$.

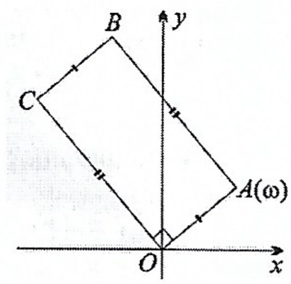
9. a. Show that $1 + i$ is a root of the polynomial equation $Q(x) = x^3 + x^2 - 4x + 6$.

   b. Hence resolve $Q(x)$ into irreducible factors over the field of real numbers.

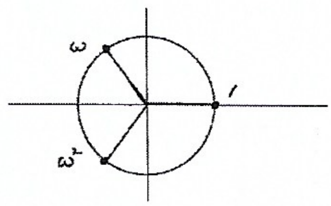
10. a. Show that $x = i$ is a root of $x^4 - x^3 - x^2 - x - 2 = 0$

    b. Hence factorise $x^4 - x^3 - x^2 - x - 2 = 0$ over the real numbers.

11. In this Argand diagram, OABC is a rectangle, with $OC = 2OA$. The vertex A corresponds to the complex number ω.
    a. What complex number corresponds to the vertex C?
    b. What complex number corresponds to the point of intersection D of the diagonals OB and AC?

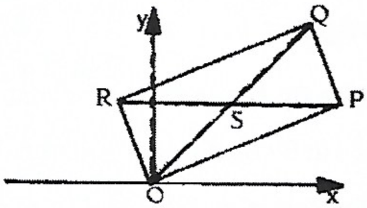


12. The vertices of an equilateral triangle are equidistant from the origin. One of its vertices is at $1 + \sqrt{3}i$. Find the complex numbers represented by the other two vertices. [HINT: What is the angle subtended by the vertices at the origin?]
13. Given that $z_1 = 1 + i$, $z_2 = 2 + 6i$ and $z_3 = -1 + 7i$, find the three possible value of z_4 so that the point representing z_1, z_2, z_3 and z_4 form a parallelogram.
14. The three cube roots of 1 are $1, \omega$ and ω^2, where $\omega = \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)$, as shown in the argand diagram.
    a. Copy the diagram and mark in the position of $1 + \omega$
    b. Hence or otherwise show that $(1 + \omega)^6 = 1$
    c. Find the smallest positive integer n such that $(1 - \omega)^n$ is real



15. Given $z_1 = 3 + 4i$ and $z_2 = -3 - 4i$
    a. Draw a neat sketch of the locus specified by $|z - z_1| = |z - z_2|$. Find the Cartesian equation of the locus of z.
    b. Show that the locus of a point represented by the complex number $z = x + iy$ and obeying the condition $|z - z_1| = 3|z - z_2|$ is a circle. Find its centre and radius.

16. P is a variable point on the line $x = 4$ and $OPQR$ is a rectangle in which the length of OP is twice the length of OR. Find the locus of S in Cartesian form, where S is the point of intersection of the diagonals.



17. a. Let $\frac{a-ib}{a+ib} = x + iy$ hence prove that $|x + iy| = 1$

b. Find the Cartesian equation of the locus of z if $\omega = \frac{z+1}{z+2i}$ and ω is purely real.

18. a. If z is a complex number such that $z + \frac{1}{z}$ is real, prove that either z is real or $|z| = 1$.

b. Given that $|z| = 1$ and $z + \frac{1}{z} = \sqrt{2}$, find the value of $z^{10} + \frac{1}{z^{10}}$

19. a. Find the roots of $z^5 + 1 = 0$, leaving your solutions in module-argument form.

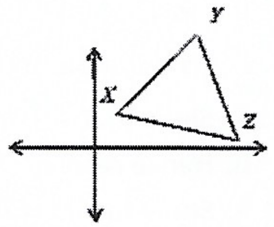
b. Prove that the solutions of $z^4 - z^3 + z^2 - z + 1 = 0$ are the non-real solutions of $z^5 + 1 = 0$.

c. By dividing through by z^2, show that if $z^4 - z^3 + z^2 - z + 1 = 0$ where $z = \cos\theta + i\sin\theta$ then $4\cos^2\theta - 2\cos\theta - 1 = 0$.

d. Hence, find the exact value of $\sec\frac{3\pi}{5}$.

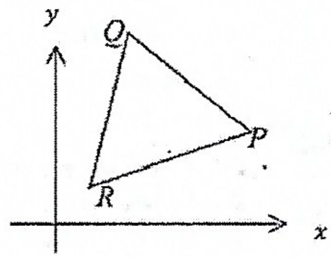
20. In the diagram, the points X, Y and Z correspond to the complex numbers x, y and z respectively. ΔXYZ is an equilateral triangle. Find the complex numbers represented by:

a. The vector OX (where $O$is the origin)
b. The vector XZ
c. The point A such that $XYAZ$ is a parallelogram
d. The point C, the centroid of ΔXYZ. (note: The centroid of a triangle is the point of intersection of the three medians.)



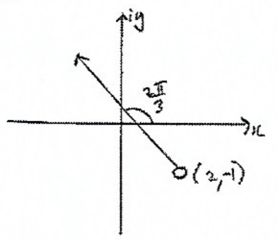
21. In the Argand diagram above, the points P, Q and R represent the complex numbers p, q, and r.

a. Given that the triangle PQR is equilateral, explain why $r - q = (\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})(q - p)$.
b. Hence, or otherwise show that $2r = (p + q) + i\sqrt{3}(q - p)$.

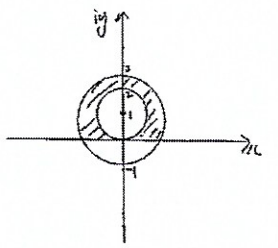


Answer

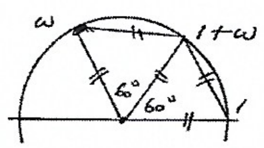
2. $8\,cis\frac{\pi}{4}$; $\pm 2\sqrt{2}cis\frac{\pi}{8}$
3. a.



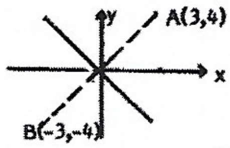
b.



4. $4xy = 1$
5. a. 2θ b. $\frac{\pi}{2} + \frac{\theta}{2}$
6. a. $\pm\frac{\pi}{2}$; b. $z_2 = (-3 + 2i)k$, k is real number
7. 36
9. b. $Q(x) = (x^2 - 2x + 2)(x + 3)$
10. b. $(x^2 + 1)(x - 2)(x + 1) = 0$
11. a. $2\omega i$ b. $\frac{1}{2}\omega(1 + 2i)$
12. -2 and $1 - \sqrt{3}i$
13. $-2 + 2i, 12i, 4$
14. a. diagram; c. $n = 6$



15. a. Locus of z is the perpendicular bisector of AB, A(3,4), B(−3, −4), it's equation $3x + 4y = 0$



b. Circle $x^2 + y^2 + \frac{15}{2}x + 10y + 25 = 0$, $C\left(-\frac{15}{4}, -5\right), r = \frac{15}{4}$

16. A straight line $2x + y = 5$
17. b. $2x + y + 2 = 0$
18. b. 0
19. a. $z = cis\left(\frac{\pi+2k\pi}{5}\right), k = 0, \dots, 4$
 d. $-\sqrt{5} - 1$
20. a. x b. $z - x$ c. $y + z - x$
 d. $\frac{x+y+z}{3}$